2 Fig. 2.1 shows an archer with a bow.

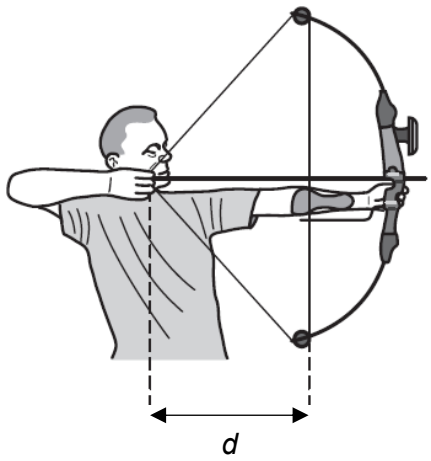

Fig. 2.1

The force *F* required to bend the bow and the corresponding displacement *d* of the arrow are measured. A plot of *F* against *d* is shown in Fig. 2.2.

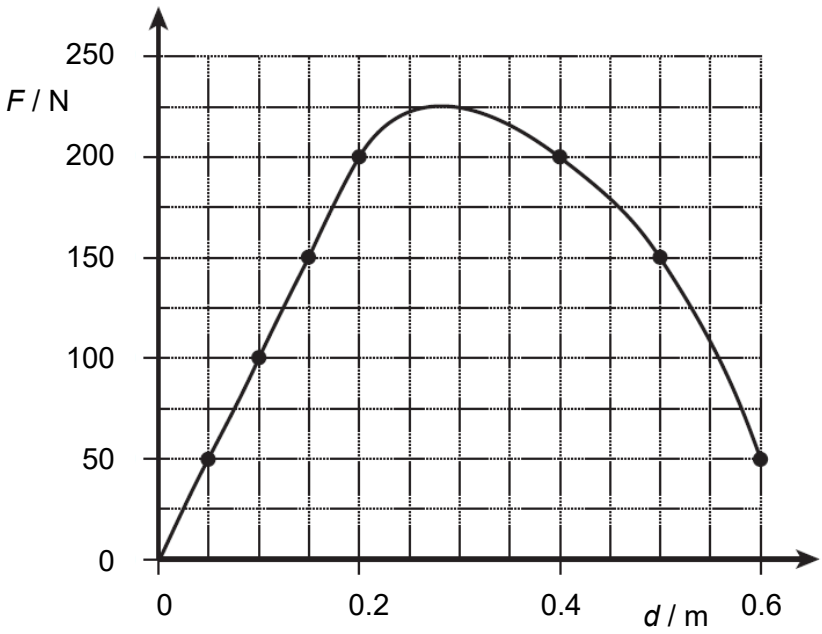

Fig. 2.2

(a) An experienced archer is able to draw an arrow further back, resulting in a greater displacement of arrow, *d*, as compared to a novice archer.

Using Fig. 2.2 and the principle of conservation of energy, suggest an advantage that this extra displacement provides.

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..... [2]

(b) **(i)** Using Fig. 2.2, estimate the elastic potential energy stored in the bow when an arrow is displaced by 0.50 m.

elastic potential energy = ...J [2]

(ii) The arrow in **(b)(i)** is then released by the archer from the bow.

Determine the maximum possible speed of the arrow which has a mass of 3.5×10^{-2} kg.

maximum possible speed = ... m s^{-1} [2]